

A flatbed scanner is usually composed of a glass pane (or platen), under which there is a bright light (often xenon or cold cathode fluorescent lamp) which illuminates the pane, and a moving optical array, whether CCD or CIS. Colour scanners typically contain three rows (arrays) of sensors with red, green and blue filters. Images to be scanned are placed face down on the glass, an opaque cover is lowered over it to exclude ambient light, and the sensor array and light source move across the pane, reading the entire area. An image is therefore visible to the charge-coupled device only because of the light it reflects. Transparent images do not work in this way, and require special accessories that illuminate them from the upper side. Many scanners offer this as an option.

Hand scanners are manual devices that are dragged across the surface of the image to be scanned. Scanning documents in this manner requires a steady hand, as an uneven scanning rate would produce distorted images—a little light on the scanner would indicate if the motion was too fast. They typically have a “start” button, which is held by the user for the duration of the scan; some switches to set the optical resolution; and a roller, which generates a clock pulse for synchronisation with the computer. Most hand scanners were monochrome, and produced light from an array of green LEDs to illuminate the image. A typical hand scanner also had a small window through which the document being scanned could be viewed.

Scanners typically read red-green-blue colour (RGB) data from the array. This data is then processed with some proprietary algorithm to correct for different exposure conditions and sent to the computer via the device’s input / output interface (usually SCSI or LPT in machines prior to the USB standard). Colour depth varies depending on the scanning array characteristics, but is usually at least 24 bits. High quality models have 48 bits or more colour depth. The other qualifying parameter for a scanner is its resolution measured in pixels per inch (ppi), more accurately referred to as samples per inch (spi). Instead of using the scanner’s true optical resolution, manufacturers refer to the